HONEYPOT HOME LAB

Reports, Findings & Technical Evidence

A controlled SSH honeypot experiment using Cowrie, Kali Linux, Windows PowerShell, and Oracle VirtualBox

Project by	Ella Faith M. Sigue
Project type	Local cybersecurity home lab
Honeypot	Cowrie 3.1.3
Client	Windows OpenSSH / PowerShell
Date of evidence	23 October 2026

Prepared for cybersecurity learning and portfolio documentation

Executive Summary

This project documents the setup and validation of a local SSH honeypot using Cowrie 3.1.3 on Kali Linux inside Oracle VirtualBox. A Windows host connected to the honeypot over a VirtualBox Host-Only network. PowerShell confirmed TCP connectivity to 192.168.56.103 on port 2222, an SSH session reached Cowrie's emulated Linux shell, and Cowrie logs recorded the connection, simulated login, and entered commands.

The test demonstrates connectivity and logging inside the local lab only. It does not show access to a school production network, a real remote server, or the underlying Kali operating system.

Key Results

Area	Observed result
Honeypot status	Cowrie reported running, PID 3578, in the captured status output.
Listening service	The twistd process was listening on 0.0.0.0:2222/TCP.
Network test	Test-NetConnection returned TcpTestSucceeded: True.
SSH interaction	The client reached Cowrie's simulated shell and ran test commands.
Logging	Cowrie recorded connection, login, shell setup, and command events.

1. Objectives and Scope

- Install and run Cowrie in a Python virtual environment on Kali Linux.
- Configure a Host-Only network so the Windows host can access the honeypot.
- Verify that Cowrie listens on TCP port 2222.
- Test TCP connectivity and SSH interaction from Windows.
- Review Cowrie logs and preserve evidence of the controlled test.

Scope and safety boundaries

The experiment was limited to a local VirtualBox lab. The Windows host was the test client, and Kali Linux hosted Cowrie. No external recipients, public honeypot exposure, real credentials, or school production systems were part of the documented test. The simulated root prompt is provided by Cowrie and is not proof of root access to Kali.

2. Lab Environment and Network Design

Component	Observed configuration
Host system	Windows with PowerShell and OpenSSH client
Virtualization	Oracle VirtualBox
Guest system	Kali Linux
Honeypot	Cowrie 3.1.3
Kali NAT address	10.0.2.15/24
Kali Host-Only address	192.168.56.103/24
Windows Host-Only address	192.168.56.1
Cowrie service	TCP 2222

The Kali VM used a NAT adapter for general outbound access and a Host-Only adapter for communication with the Windows host. The Host-Only address was the target for the test. Cowrie listened on port 2222 rather than the default SSH port 22.

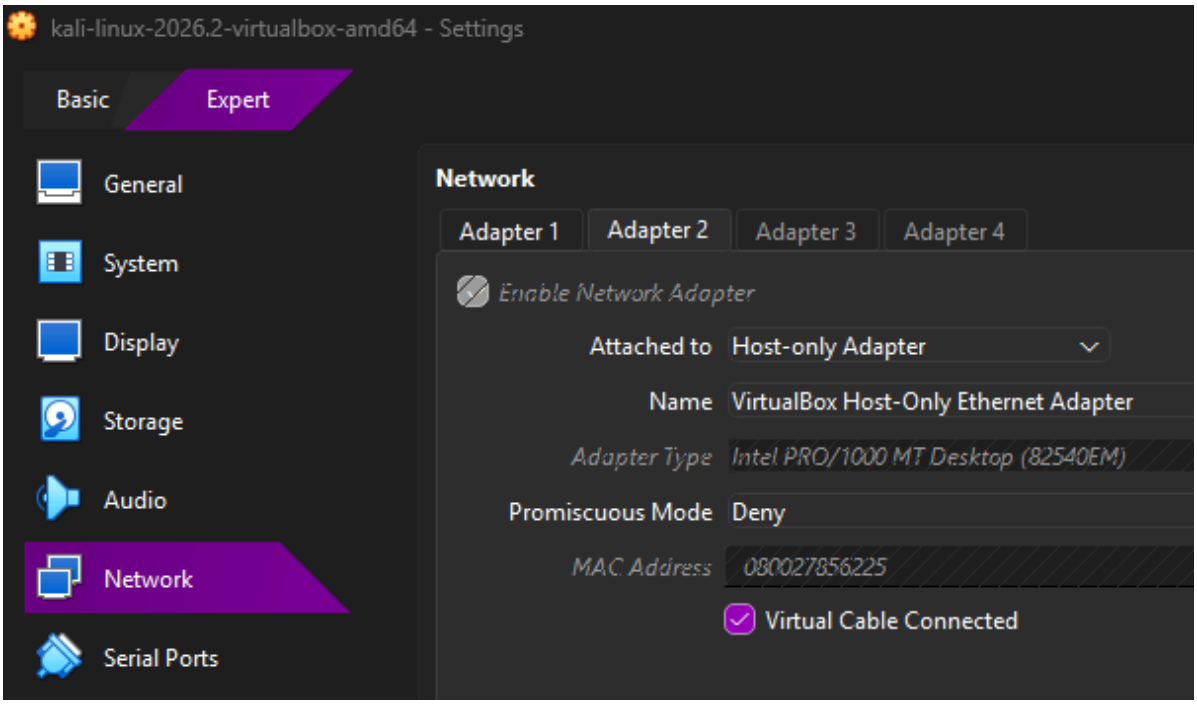


Figure 1. VirtualBox Adapter 2 configured as a Host-Only Adapter.

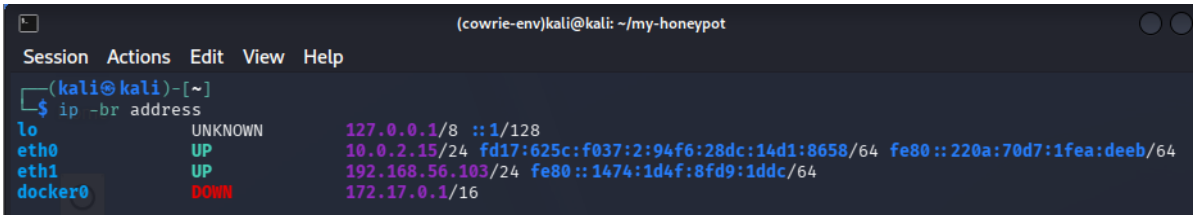


Figure 2. Kali Linux network interfaces, including eth1 at 192.168.56.103.

3. Service Status and Port Verification

Cowrie was checked using its command-line status utility. The ss command was used to confirm that a process was listening on TCP port 2222. The captured output showed Cowrie running as PID 3578 and the associated twistd process listening on 0.0.0.0:2222.

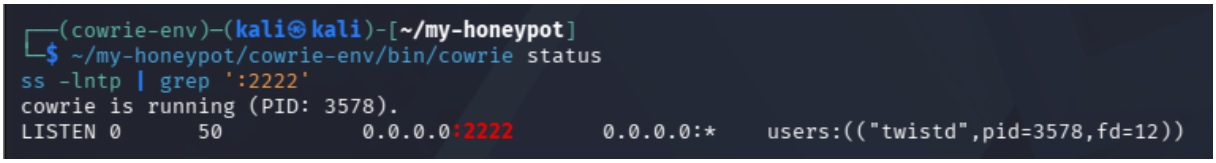


Figure 3. Cowrie running and the twistd process listening on TCP port 2222.

4. Windows Connectivity Test

From Windows PowerShell, Test-NetConnection was run against the Kali Host-Only IP and Cowrie port:

```
Test-NetConnection 192.168.56.103 -Port 2222
```

The output reported TcpTestSucceeded : True, confirming that the Windows host established a TCP connection to the service at the time of testing.

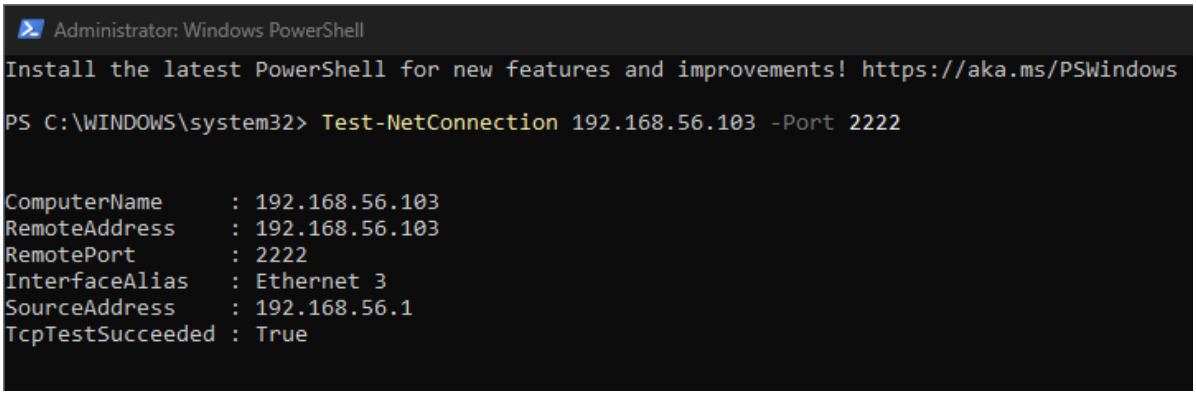


Figure 4. PowerShell TCP connectivity test returned TcpTestSucceeded: True.

5. SSH Session and Simulated Shell Interaction

The client connected using the explicit Cowrie port:

```
ssh -p 2222 root@192.168.56.103
```

The session displayed a Debian-style banner and a prompt resembling root@svr04:~#. Commands including whoami, pwd, ls, mkdir Glenn, and ls -la were entered. Cowrie emulates a Linux server, so the displayed root identity and filesystem are part of the simulated environment rather than the actual Kali operating system.

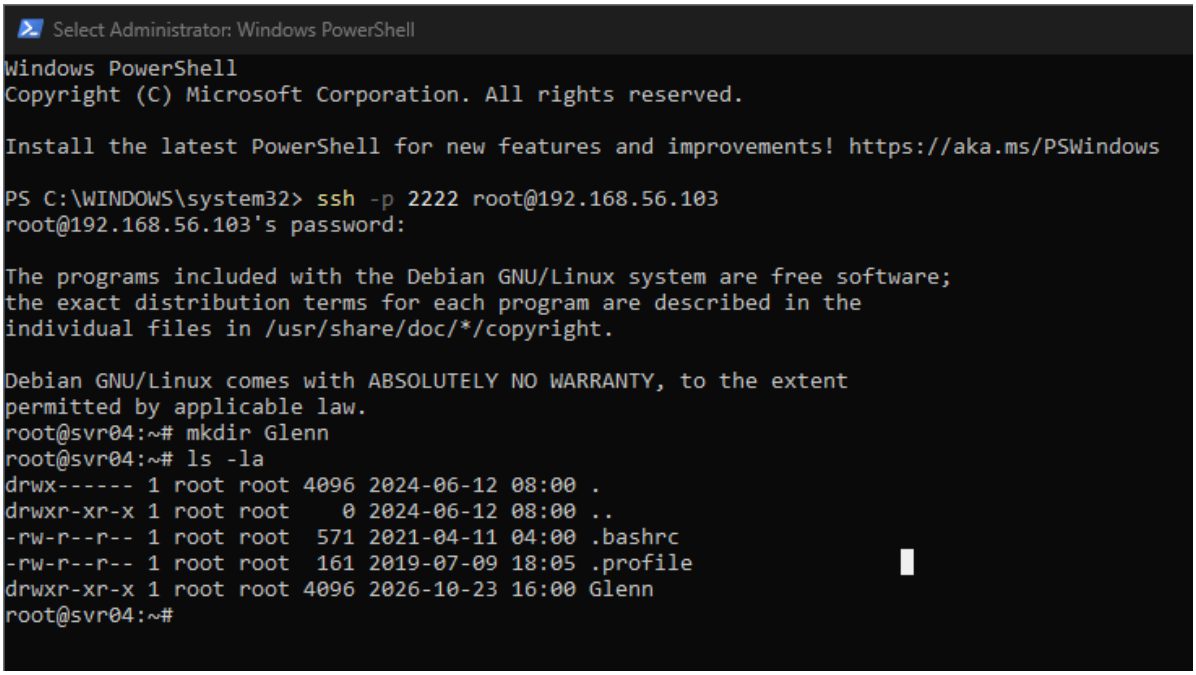


Figure 5. Windows SSH client interacting with Cowrie's simulated shell.

6. Log Review and Findings

The primary Cowrie log was reviewed at ~/my-honeypot/var/log/cowrie/cowrie.log. The excerpts showed service startup, incoming connections from 192.168.56.1, the Windows OpenSSH client identification, an accepted simulated login, shell initialization, and command events.

Log evidence	Interpretation
New connection	Cowrie logged a connection from Windows host 192.168.56.1 to Kali 192.168.56.103:2222.
SSH client identification	The log identified the client as OpenSSH_for_Windows_9.5.
Simulated login	The log recorded login attempt [root/kali] succeeded. This is a Cowrie-emulated login, not authentication to the actual Kali system.
Shell and commands	The log recorded shell setup and commands such as whoami, pwd, ls, mkdir Glenn, and ls -la.
Connection termination	Some sessions ended after a timeout or disconnect, which is consistent with short controlled test sessions.

```
┌──(cowrie-env)─(kali㉿kali)-[~/my-honeypot]
└─$ tail -n 50 ~/my-honeypot/var/log/cowrie/cowrie.log
2026-10-23T15:45:18.838027Z [cowrie.core.config#info] Reading configuration from ['etc/cowrie.cfg']
2026-10-23T15:45:19.155886Z [cowrie.plugin#info] Python Version 3.14.8 (main, Oct  1 2026, 19:55:55) [GCC 16.2.0]
2026-10-23T15:45:19.155907Z [cowrie.plugin#info] Twisted Version 26.4.0
2026-10-23T15:45:19.155914Z [cowrie.plugin#info] Cowrie Version 3.1.3
2026-10-23T15:45:19.155918Z [cowrie.plugin#info] Sensor UUID: 14dfb6d4-ceb4-11f1-a19c-0800275a87bc
2026-10-23T15:45:19.157477Z [cowrie.core.output#info] Loaded output engine: jsonlog
2026-10-23T15:45:19.158775Z [twisted.scripts._twistd_unix.UnixAppLogger#info] twistd 26.4.0 (/home/kali/my-honeypot/
cowrie-env/bin/python 3.14.8) starting up.
2026-10-23T15:45:19.158815Z [twisted.scripts._twistd_unix.UnixAppLogger#info] reactor class: twisted.internet.epollr
eactor.EPollReactor.
2026-10-23T15:45:19.162247Z [-] CowrieSSHFactory starting on 2222
2026-10-23T15:45:19.162710Z [cowrie.ssh.factory.CowrieSSHFactory#info] Starting factory <cowrie.ssh.factory.CowrieSS
HFactory object at 0×7fbe3a6a9a90>
2026-10-23T15:45:19.203855Z [cowrie.ssh.factory.CowrieSSHFactory#info] Ready to accept SSH connections
2026-10-23T15:45:43.247321Z [ssh,a54591ed76ad,192.168.56.1] New connection: 192.168.56.1:49883 (192.168.56.103:2222)
 [session: a54591ed76ad]
2026-10-23T15:45:43.247856Z [HoneyPotSSHTransport,0,192.168.56.1] connection lost
2026-10-23T15:45:43.248036Z [ssh,a54591ed76ad,192.168.56.1] Connection lost after 0 milliseconds
2026-10-23T15:46:13.574601Z [ssh,92efdf652b6e,192.168.56.1] New connection: 192.168.56.1:49976 (192.168.56.103:2222)
 [session: 92efdf652b6e]
2026-10-23T15:46:13.575153Z [ssh,92efdf652b6e,192.168.56.1] Remote SSH version: SSH-2.0-OpenSSH_for_Windows_9.5
2026-10-23T15:46:13.575712Z [ssh,92efdf652b6e,192.168.56.1] SSH client hassh fingerprint: 701158e75b508e76f0410d5d22
ef9df0
2026-10-23T15:46:28.445322Z [ssh,92efdf652b6e,192.168.56.1] login attempt [root/kali] succeeded
2026-10-23T15:46:28.445485Z [HoneyPotSSHTransport,1,192.168.56.1] Initialized emulated server as architecture: linux
-x64-lsb
2026-10-23T15:46:28.446994Z [HoneyPotSSHTransport,1,192.168.56.1] channel open
2026-10-23T15:46:28.448943Z [SSHChannel session (0) on SSHService b'ssh-connection' on HoneyPotSSHTransport,1,192.16
8.56.1] Handling pty request: b'xterm-256color' (51, 102, 640, 480)
2026-10-23T15:46:28.449105Z [ssh,92efdf652b6e,192.168.56.1] Terminal Size: 102 51
2026-10-23T15:46:28.449291Z [SSHChannel session (0) on SSHService b'ssh-connection' on HoneyPotSSHTransport,1,192.16
8.56.1] Getting shell
2026-10-23T15:49:00.120409Z [ssh,92efdf652b6e,192.168.56.1] CMD: whoami
2026-10-23T15:49:00.123162Z [HoneyPotSSHTransport,1,192.168.56.1] Command found: whoami
2026-10-23T15:49:08.485781Z [ssh,92efdf652b6e,192.168.56.1] CMD: pwd
2026-10-23T15:49:08.487277Z [HoneyPotSSHTransport,1,192.168.56.1] Command found: pwd
2026-10-23T15:49:14.598085Z [ssh,92efdf652b6e,192.168.56.1] CMD: ls
2026-10-23T15:49:14.600588Z [HoneyPotSSHTransport,1,192.168.56.1] Command found: ls
2026-10-23T15:51:28.542533Z [cowrie.ssh.transport.HoneyPotSSHTransport#info] Timeout reached in HoneyPotSSHTransport
2026-10-23T15:51:28.543478Z [ssh,92efdf652b6e,192.168.56.1] Closing TTY Log: var/lib/cowrie/tty/48c28e634a269aa42468
dcad54bba62a7d4f9553b245807e495cba67d2b07e43 after 300094 milliseconds
2026-10-23T15:51:28.543599Z [HoneyPotSSHTransport,1,192.168.56.1] avatar root logging out
2026-10-23T15:51:28.543755Z [HoneyPotSSHTransport,1,192.168.56.1] connection lost
2026-10-23T15:51:28.544004Z [ssh,92efdf652b6e,192.168.56.1] Connection lost after 314969 milliseconds
2026-10-23T15:57:46.910246Z [ssh,90e3e616b342,192.168.56.1] New connection: 192.168.56.1:58974 (192.168.56.103:2222)
 [session: 90e3e616b342]
2026-10-23T15:57:46.913128Z [HoneyPotSSHTransport,2,192.168.56.1] connection lost
2026-10-23T15:57:46.913339Z [ssh,90e3e616b342,192.168.56.1] Connection lost after 3 milliseconds
2026-10-23T15:59:09.389898Z [ssh,450def877fdc,192.168.56.1] New connection: 192.168.56.1:51519 (192.168.56.103:2222)
 [session: 450def877fdc]
```

Figure 6. Cowrie log excerpt showing startup, connection, simulated login, and initial commands.

```
2026-10-23T15:45:43.247321Z [ssh,a54591ed76ad,192.168.56.1] New connection: 192.168.56.1:49883 (192.168.56.103:2222)
 [session: a54591ed76ad]
2026-10-23T15:45:43.247856Z [HoneyPotSSHTransport,0,192.168.56.1] connection lost
2026-10-23T15:45:43.248036Z [ssh,a54591ed76ad,192.168.56.1] Connection lost after 0 milliseconds
2026-10-23T15:46:13.574601Z [ssh,92efdf652b6e,192.168.56.1] New connection: 192.168.56.1:49976 (192.168.56.103:2222)
 [session: 92efdf652b6e]
2026-10-23T15:46:13.575153Z [ssh,92efdf652b6e,192.168.56.1] Remote SSH version: SSH-2.0-OpenSSH_for_Windows_9.5
2026-10-23T15:46:13.575712Z [ssh,92efdf652b6e,192.168.56.1] SSH client hassh fingerprint: 701158e75b508e76f0410d5d22
ef9df0
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8.56.1] Handling pty request: b'xterm-256color' (51, 102, 640, 480)
2026-10-23T15:46:28.449105Z [ssh,92efdf652b6e,192.168.56.1] Terminal Size: 102 51
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2026-10-23T15:49:08.485781Z [ssh,92efdf652b6e,192.168.56.1] CMD: pwd
2026-10-23T15:49:08.487277Z [HoneyPotSSHTransport,1,192.168.56.1] Command found: pwd
2026-10-23T15:49:14.598085Z [ssh,92efdf652b6e,192.168.56.1] CMD: ls
2026-10-23T15:49:14.600588Z [HoneyPotSSHTransport,1,192.168.56.1] Command found: ls
2026-10-23T15:51:28.542533Z [cowrie.ssh.transport.HoneyPotSSHTransport#info] Timeout reached in HoneyPotSSHTransport
2026-10-23T15:51:28.543478Z [ssh,92efdf652b6e,192.168.56.1] Closing TTY Log: var/lib/cowrie/tty/48c28e634a269aa42468
dcad54bba62a7d4f9553b245807e495cba67d2b07e43 after 300094 milliseconds
2026-10-23T15:51:28.543599Z [HoneyPotSSHTransport,1,192.168.56.1] avatar root logging out
2026-10-23T15:51:28.543755Z [HoneyPotSSHTransport,1,192.168.56.1] connection lost
2026-10-23T15:51:28.544004Z [ssh,92efdf652b6e,192.168.56.1] Connection lost after 314969 milliseconds
2026-10-23T15:57:46.910246Z [ssh,90e3e616b342,192.168.56.1] New connection: 192.168.56.1:58974 (192.168.56.103:2222)
 [session: 90e3e616b342]
2026-10-23T15:57:46.913128Z [HoneyPotSSHTransport,2,192.168.56.1] connection lost
2026-10-23T15:57:46.913339Z [ssh,90e3e616b342,192.168.56.1] Connection lost after 3 milliseconds
2026-10-23T15:59:09.389898Z [ssh,450def877fdc,192.168.56.1] New connection: 192.168.56.1:51519 (192.168.56.103:2222)
 [session: 450def877fdc]
2026-10-23T15:59:09.390199Z [ssh,450def877fdc,192.168.56.1] Remote SSH version: SSH-2.0-OpenSSH_for_Windows_9.5
2026-10-23T15:59:09.391794Z [ssh,450def877fdc,192.168.56.1] SSH client hassh fingerprint: 701158e75b508e76f0410d5d22
ef9df0
2026-10-23T15:59:12.801333Z [ssh,450def877fdc,192.168.56.1] login attempt [root/kali] succeeded
2026-10-23T15:59:12.801479Z [HoneyPotSSHTransport,3,192.168.56.1] Initialized emulated server as architecture: linux
-x64-lsb
2026-10-23T15:59:12.805536Z [HoneyPotSSHTransport,3,192.168.56.1] channel open
2026-10-23T15:59:12.806843Z [SSHChannel session (0) on SSHService b'ssh-connection' on HoneyPotSSHTransport,3,192.16
8.56.1] Handling pty request: b'xterm-256color' (62, 117, 640, 480)
2026-10-23T15:59:12.806971Z [ssh,450def877fdc,192.168.56.1] Terminal Size: 117 62
2026-10-23T15:59:12.807066Z [SSHChannel session (0) on SSHService b'ssh-connection' on HoneyPotSSHTransport,3,192.16
8.56.1] Getting shell
2026-10-23T16:00:21.958098Z [ssh,450def877fdc,192.168.56.1] CMD: mkdir Glenn
2026-10-23T16:00:21.960108Z [HoneyPotSSHTransport,3,192.168.56.1] Command found: mkdir Glenn
2026-10-23T16:00:53.275630Z [ssh,450def877fdc,192.168.56.1] CMD: ls -la
2026-10-23T16:00:53.278059Z [HoneyPotSSHTransport,3,192.168.56.1] Command found: ls -la
```

Figure 7. Additional Cowrie log excerpt showing later SSH interaction and commands.

7. Challenges and Troubleshooting

A connection attempt using ssh root@192.168.56.103 failed because SSH defaults to TCP port 22. Cowrie was configured to listen on TCP port 2222, so the correct command was ssh -p 2222 root@192.168.56.103. The PowerShell test also needed to specify port 2222. This distinction was verified by a successful TCP test and the Cowrie logs.

A separate connection attempt that ended immediately was also present in the logs. The later session progressed through SSH negotiation, simulated login, shell initialization, and command logging.

8. Skills Demonstrated

- Linux command-line administration and Python virtual-environment use.
- VirtualBox network configuration using NAT and Host-Only networking.
- TCP port verification with ss and PowerShell Test-NetConnection.
- SSH client usage and troubleshooting of non-default service ports.
- Review and interpretation of honeypot event logs.
- Evidence collection, technical documentation, and accurate reporting of lab limitations.

9. Conclusion

The local honeypot lab achieved its main objective. Cowrie was running and listening on TCP port 2222, Windows successfully reached the service through the Host-Only network, an SSH session interacted with the simulated shell, and Cowrie recorded relevant activity in its logs. The test provides repeatable evidence of local connectivity and honeypot logging. It does not demonstrate access to a school production network or the underlying Kali host.

10. Evidence and Reproducibility Commands

Purpose	Command
Kali interface addresses	ip -br address
Cowrie status	~/my-honeypot/cowrie-env/bin/cowrie status
Listening port	ss -lntp | grep ':2222'
Windows TCP test	Test-NetConnection 192.168.56.103 -Port 2222
SSH to Cowrie	ssh -p 2222 root@192.168.56.103
View recent log entries	tail -n 50 ~/my-honeypot/var/log/cowrie/cowrie.log
Follow live logs	tail -f ~/my-honeypot/var/log/cowrie/cowrie.log

Evidence handling note: retain the original log file and unedited screenshots. If a report image contains a test password or other sensitive detail, redact only the copy used in the report and keep the original evidence securely.